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A DISK-BASED DYNAMICAL CONSTRAINT ON THE MASS OF THE YOUNG BINARY DQ TAU

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ABSTRACT

We present spatially and spectrally resolved Atacama Large Millimeter/submillimeter Array (ALMA) observations of gas and dust in the disk orbiting the pre-main sequence binary DQ Tau. By forward-modeling the disk velocity field traced by CO J=2−1 line emission, we infer the mass of the central binary, $M_* = 1.2 \pm 0.2\,M_\odot$, a new dynamical measurement that is independent of stellar evolutionary models. Assuming the disk and binary are co-planar within $\sim 2^\circ$, this disk-based binary mass measurement is in excellent agreement with constraints from radial velocity monitoring of the combined stellar spectra. These ALMA results are also compared with the standard approach of estimating masses from the location of the binary in the Hertzsprung-Russell diagram, using several common pre-main sequence model grids. These models predict stellar masses that are consistent with the dynamical mass measurement. These same models consistently predict an age of 5 Myr. As ALMA accumulates comparable data for large samples of pre-main sequence stars, the methodology employed here to extract a dynamical mass from the disk rotation curve should prove extraordinarily useful for efforts to characterize the fundamental parameters of early stellar evolution.

Keywords: protoplanetary disks – stars: fundamental parameters – stars: pre-main sequence – stars: individual (DQ Tau)

1. INTRODUCTION

Theoretical models for pre-main sequence (pre-MS) stellar evolution are fundamental tools for learning about star and planet formation. But the accuracy of such models – especially at young ages – is unclear, due to our limited understanding of some complex physical effects like accretion (e.g., Baraffe et al. 2009) or magnetic fields and convection (e.g., Feiden & Chaboyer 2013). While such issues are being explored theoretically, robust observational constraints on key stellar parameters can be used to help guide improvements to the models. Most important are dynamical constraints on stellar masses, M_* (e.g., see Hillenbrand & White 2004). Usually these are determined from the orbital motions of binary systems (Stassun et al. 2014), but they could increasingly be measured for *single* stars based on the rotation of their associated gas disks (e.g., Simon et al. 2000).

We are interested in comparing the constraints from these two approaches, to illuminate (and quantify) any associated systematic problems in the inference of M_*. To do that, we have targeted the few roughly equal-mass double-lined spectroscopic binaries that host circumbinary disks, including V4046 Sgr (Rosenfeld et al. 2012b) and AK Sco (Czekala et al. 2015). In both cases, excellent agreement (to within $\sim$1%) is found between the estimates of M_* from radial velocity monitoring of the stars and the tomographic reconstruction of CO gas in the disk. The confluence of these measurements also indicates that the disk and binary orbital planes are well-aligned (within 1–2°). Moreover, the predictions of theoretical pre-MS faithfully reproduce these results in these two examples. However, these results are perhaps not surprising, since both V4046 Sgr and AK Sco are relatively old (10 and 18 Myr) and massive (1.8 and 2.5 $M_\odot$ in each system) and the models should be more robust in that range of age and mass. An important supplementary test would employ a cooler and younger binary.

In those respects, DQ Tau is an exemplary target. DQ Tau is an equal-mass ($q = 0.97 \pm 0.15$) double-lined spectroscopic binary with a period of 15.8 days and a notably eccentric orbit (Mathieu et al. 1997). It has a composite spectral type of $\sim$M0–M1 (e.g., Herbig 1977; Herczeg & Hillenbrand 2014) and is located in the nearby ($d \sim 140$ pc) and relatively young ($\sim$1–3 Myr) Taurus clouds. This system exhibits broadband optical brightesning (Mathieu et al. 1997), emission line variations (Basri et al. 1997), and mm/radio enhancements (Salter et al. 2010) near peri-astron, which have been associated with both pulsed accretion and reconnection events from colliding magnetospheres. The DQ Tau system hosts a circumbinary disk with substantial (sub-)millimeter continuum emission from dust (Beckwith et al. 1990; Andrews & Williams 2005; Guilloteau et al. 2011). There is recent evidence for molecular gas in rotation around the central binary host (Williams & Best 2014), although there is non-negligible contamination from the local molecular cloud environment (Guilloteau et al. 2013).

We present new observations of molecular gas in the DQ Tau circumbinary disk made with the Atacama Large Millimeter/Submillimeter Array (ALMA), and use them to place a dynamical constraint on the total mass of the DQ Tau binary. We also provide an updated orbital solution for the DQ Tau binary based on long-term radial velocity monitoring. Section 2 presents the data and its calibration. Section 3 describes our modeling of the gas disk velocity field, presents an update of the original Mathieu et al. (1997) orbital solution, and highlights the

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key results. And Section 4 discusses these results in the contexts of other young circumbinary disk systems, the predictions of pre-MS evolution models, and the utility of the disk-based dynamical mass technique.

2. OBSERVATIONS AND DATA REDUCTION

2.1. *Millimeter Interferometry*

ALMA observed the DQ Tau system on 2015 May 24, using 34 of its 12 m antennas with separations ranging from 21 to 540 m. The observations were configured with the same spectral setup as in Czekala et al. (2015), employing the Band 6 receivers to cover the CO, ^{13}CO, and C^{18}O J=2−1 transitions in 61, 61, and 122 kHz channels, respectively, as well as the adjacent continuum (at 232 GHz, or 1.3 mm). The nearby quasar J0510+1800 (6° separation) was observed regularly to monitor variations in the complex gain response of the interferometer. The bright quasar J0423−0120 was also observed to calibrate the bandpass behavior and absolute flux levels. The total on-source integration time was ∼28 minutes. The visibilities were calibrated using standard techniques with the `CASA` software package (v4.3.1). After a phase and amplitude self-calibration based on the bright continuum, the spectral line visibilities were time-averaged (to 30 s intervals) and continuum-subtracted.

Images of the continuum and spectral line data were created by Fourier inverting the calibrated visibilities (assuming a Briggs robust weighting parameter of 0.5, to balance S/N and resolution), deconvolving with the `CLEAN` algorithm, and restoring with a synthesized beam with FWHM = $0\farcs8 \times 0\farcs6$ (at P.A. = 145°). The continuum image shows a bright, marginally resolved source with a peak intensity of ∼68 mJy beam^{-1} and integrated flux density of 79 mJy. The RMS noise level is 70 μJy beam^{-1} (the peak S/N is ∼1000; the map sensitivity is clearly limited by dynamic range). All of this emission is expected to be from dust; the peri-apse continuum enchanement noted by Salter et al. (2010) is not present at the observed orbital phase ($\phi = 0.71$). But the focus here is on the emission from the CO spectral lines.

The ^{12}CO (hereafter CO) and ^{13}CO J=2−1 transitions were imaged in 0.1 km s^{-1}-wide channels, and reach an RMS noise level of 8 mJy beam^{-1} in each channel. Line emission from these transitions is detected over a ∼7 km s^{-1}-wide velocity range, exhibiting the classical morphological pattern of Keplerian rotation. The peak S/N is 35 for CO, but only 7 for ^{13}CO. Both of these transitions show considerable contamination from the local molecular cloud material, affecting a 2 km s^{-1}-wide span slightly blueshifted from the systemic velocity. The C^{18}O J=2−1 transition was imaged in 0.2 km s^{-1}-wide channels, with an RMS of ∼4 mJy beam^{-1} in each, but the line is only marginally detected (S/N∼3) in a few of these channels. Given the line intensities, our focus will be on an interpretation of the CO emission. The CO channel maps are shown in the top portion of Figure 1.

2.2. *Optical Spectroscopy*

For this work we have made use of three sets of spectroscopic observations for DQ Tau, including material also used by Mathieu et al. (1997). The first set consists of 30 spectra obtained at the Harvard-Smithsonian Center for Astrophysics (CfA) between 1984 and 2005 with two similar instruments equipped with intensified photon-counting Reticon detectors, as described in more detail by the above authors. These spectrographs are no longer in operation. A subset of 23 of these spectra were used by Mathieu et al. (1997); we have reanalyzed all 30 of them here with improved techniques. These single-order spectra (45 Å centered around the Mg I b triplet near 5190 Å) have relatively low S/N, ranging from 6 to 16 per resolution element of 8.5 km s^{-1}. A second set of 22 observations consists of radial velocity differences measured from spectra also described by Mathieu et al. (1997) and collected with instruments at the Lick Observatory, the Keck Observatory, and the McDonald Observatory. Finally, more recently (2013 October to December) we obtained three additional spectra of DQ Tau at the CfA for a different purpose, with the 1.5 m Tillinghast reflector at the Fred L. Whipple Observatory on Mount Hopkins (AZ). For this we used the bench-mounted TRES instrument (Fűrész 2008) that delivers a resolving power of $R \approx 44,000$ in 51 echelle orders spanning the wavelength range 3900–9100 Å. These three spectra have signal-to-noise ratios in the Mg I b region of 16, 26, and 23 per resolution element of 6.8 km s^{-1}.

Radial velocities (RVs) for each component of DQ Tau were measured from all of the CfA spectra using the two-dimensional cross-correlation technique TODCOR (Zucker & Mazeh 1994), with templates taken from a library of synthetic spectra based on PHOENIX model atmospheres (see Husser et al. 2013) computed for the appropriate instrumental resolution. Because the spectral lines of the two stars are almost always at least partially blended in these spectra, the choice of templates is rather critical in order to obtain accurate RVs, particularly regarding the rotational broadening. Based on indications from the work of Mathieu et al. (1997) that the mass ratio is close to unity, we assumed the stars have the same temperature. We also adopted surface gravities of $\log g = 4.5$ for both components following Bary & Petersen (2014), as well as solar metallicity. We experimented with a variety of template combinations, and found the best match to the DQ Tau spectra for a temperature of 4200 K and $v \sin i$ values of 14 and 11 km s^{-1} for the primary and secondary, respectively. These latter values are similar to the measurements of Nguyen et al. (2012), who obtained 14.7 ± 1.6 and 11.3 ± 0.7 km s^{-1}. The temperature corresponds formally to a spectral type of K6 or K7, although $T_{\rm eff}$ is degenerate with the surface gravity in our procedure because of the short wavelength coverage of the Reticon spectra from which we made these determinations. However, the RVs are unaffected by this degeneracy so long as the temperature for the templates is chosen to provide the optimal match to the spectra for a given $\log g$ value. Rotational broadening has a much larger effect on the velocities in this case because of the heavy line blending, and we believe our fine-tuning of this parameter for both stars is the reason we are able to derive meaningful velocities from all 23 of the spectra used by Mathieu et al. (1997) (in addition to the other 10 from CfA used here). Their procedures only allowed them to derive separate velocities for 14 of their least blended spectra, the rest providing only an upper limit of 18 km s^{-1} on the velocity separation between the

primary and secondary. We list our new radial-velocity measurements from all CfA spectra in Table 1. In addition to the velocities, we also measured a flux ratio between the stars of $\ell_2/\ell_1 = 0.85 \pm 0.08$ at the mean wavelength of the CfA spectra (5190 Å).

The velocity zero-point of the Reticon observations was monitored each night by means of dusk and dawn exposures of the twilight sky, and small run-to-run corrections were applied in the manner described by Latham (1992). For TRES the zero point was monitored by observing IAU velocity standards each night. All velocities were placed on the same system, and the measurements listed in Table 1 include all corrections.

3. ANALYSIS AND RESULTS

3.1. *CO Disk Modeling and Results*

We analyze the CO J=2−1 line emission using the framework detailed in Czekala et al. (2015) and Rosenfeld et al. (2012b). Briefly, we forward-model the ALMA visibilities using a parametric description of the disk structure (densities, temperatures, and velocities). For any set of model parameters, we calculate the excitation conditions in the disk assuming local thermodynamic equilibrium. We then use the `RADMC-3D` radiative transfer code[5] to ray-trace spectral images, which are Fourier transformed and sampled at the same spatial frequencies as the data. A χ^2 likelihood function is used to assess the fit quality. The posterior parameter-space is explored with a Markov Chain Monte Carlo (MCMC) algorithm.[6]

The model parameters can be catalogued into four groups. The first group includes parameters that describe the CO gas densities. We assume the standard Lynden-Bell & Pringle (1974) similarity solution describes the radial surface density profile of the gas, which is described by a gradient γ, a characteristic radius r_c, and a normalization that we cast in terms of the total mass $M_{\rm CO}$. For computational expediency, we fix $\gamma = 1$. The second group describes the gas temperatures. We simplify the scenario by assuming a vertically isothermal structure, with a radial power-law temperature profile that has normalization T_{10} (the temperature at 10 AU) and index q. This thermal structure is employed in calculating the vertical density distribution, assuming hydrostatic equilibrium. The third group of parameters sets the projected velocity field of the gas, presumed to be in Keplerian rotation. It includes the central binary mass M_*, the disk inclination i_d and position angle φ, and a systemic velocity $v_{\rm sys}$. Non-thermal line broadening is permitted with a line-width ξ added in quadrature to the normal themal contribution. In our convention, $i_d = 0$ is a face-on disk with the angular momentum vector pointed towards the observer, $i_d = 90$ is edge-on disk, and $i = 180$ is a face on disk, but with the disk angular momentum vector pointed away from the observer. Position angle φ is defined by the projection of the angular momentum axis onto the sky measured in degrees East of North. And the fourth group comprises a set of nuisance parameters, including position offsets from the observed phase center (Δ_α, Δ_δ), and the distance d.

[5] ita.uni-heidelberg.de/~dullemond/software/radmc-3d/

[6] The code used to perform the analysis described here is open source and freely available under an MIT license at https://github.com/iancze/JudithExcalibur

We explore the 12-dimensional posterior-space with MCMC, employing uniform (uninformative) priors on all parameters except for i_d and d. We adopt a geometrical prior on the disk inclination to account for the fact that the edge-on geometry occurs more frequently than a pole-on geometry, $p(i_d) = \sin(i_d)/2$. We adopt a conservative distance prior in the form of a Gaussian distribution with a mean of 145 pc and a width (σ) of 20 pc, meant to represent the range of possible distances to members of the Taurus clouds. In this analysis, we exclude 23 channels that show clear evidence of molecular cloud contamination from the likelihood calculations (these are marked in Fig. 1). The resulting inferences on the model parameters are list in Table 2. A comparison of the data with the model is shown in the form of channel maps in Figure 1. As has been demonstrated clearly in previous work (e.g., Simon et al. 2000; Rosenfeld et al. 2012b), the density- and temperature-related parameters have negligible impact on an inference of the host mass. The key parameters of interest here are M_* and i_d (as well as d). The M_*–i_d and M_*-d degeneracies are shown in Figure 2.

We infer a mass of $1.4^{+0.8}_{-0.4}\,M_\odot$ for the DQ Tau binary, marginalized over the uncertainty contained in our distance prior. This can be expressed in a distance-independent manner as $M_*/d = 0.010^{+0.005}_{-0.003}\,M_\odot$ pc^{-1}; the formal uncertainty on M_* is ∼35% if the distance is known exactly. The precision of this constraint is significantly poorer than for most disk-based dynamical mass measurements, due to the unfortunate combination of a relatively face-on viewing geometry ($i_d \approx 160°$) and the cloud contamination around the systemic velocity.

3.2. *An Updated Orbital Solution*

The spectroscopic orbital solution by Mathieu et al. (1997) used their 14 pairs of CfA RVs along with the velocities measured from the Lick, Keck, and McDonald observatories (hereafter the 'LKM' set). Because of difficulties in maintaining a consistent velocity zero point from night to night and instrument to instrument, the latter data were originally derived only as velocity *differences* between the primary and secondary, rather than individual velocities for each star. In order to be able to incorporate these LKM data into a conventional double-lined orbital solution, and at the same time to tie those observations to the CfA zero point, Mathieu et al. (1997) elected to estimate primary and secondary "measurements" from each velocity difference. They did this by assuming a mass ratio of unity, which was near their value of $M_2/M_1 = 0.97 \pm 0.15$ from an initial fit based only on their 14 CfA pairs, and adopting a center-of-mass velocity of $\gamma = 22$ km s^{-1}, from the same initial solution. They then combined all of the measurements into a final fit, but were careful to note that both γ and the velocity semiamplitudes K_1 and K_2 from that fit are biased and should be ignored in favor of the values from the CfA-only solution, and similarly with the minimum masses and projected semimajor axes, which depend on the semiamplitudes.

For this work we have preferred to incorporate the LKM velocity differences directly into the fit in their original form. We therefore reconstructed the original velocity differences trivially from the primary and secondary "measurements" reported by Mathieu et al. (1997) in

Table 1
Heliocentric radial velocity measurements of DQ Tau from CfA.

HJD (2,400,000+)	Orbital phase[a]	RV_1 (km s^{-1})	σ_1 (km s^{-1})	$(O-C)_1$ (km s^{-1})	RV_2 (km s^{-1})	σ_2 (km s^{-1})	$(O-C)_2$ (km s^{-1})
45982.0293	0.1435	34.76	3.93	−2.71	12.44	2.53	+1.75
46389.8446	0.9520	−2.93	3.61	−0.57	56.13	2.32	+2.89
46745.8122	0.4794	37.11	7.72	+6.50	13.90	4.97	−4.12
47073.9364	0.2446	37.38	3.56	+0.49	15.01	2.29	+3.69
47075.9910	0.3747	34.73	3.35	+0.97	17.06	2.16	+2.41
47078.0084	0.5023	31.64	3.23	+1.79	20.70	2.08	+1.87
47127.8232	0.6549	26.07	4.56	+2.22	24.40	2.94	−0.84
47128.8081	0.7172	23.51	5.11	+2.88	27.30	3.29	−1.38
47159.6132	0.6667	20.10	4.96	−3.19	25.73	3.19	−0.11
47159.6318	0.6679	17.31	4.46	−5.92	27.86	2.87	+1.96
47427.9265	0.6468	27.28	3.19	+3.06	23.67	2.06	−1.17
47546.6122	0.1579	36.99	5.66	−0.66	7.64	3.64	−2.86
47546.6272	0.1588	33.17	5.28	−4.48	10.94	3.40	+0.44
47546.6272	0.1588	35.11	3.15	−2.54	9.87	2.02	−0.63
47546.6423	0.1598	35.24	6.15	−2.42	10.05	3.96	−0.44
47789.9412	0.5569	28.81	2.92	+0.89	19.93	1.88	−0.96
47840.8048	0.7758	12.43	3.01	−4.45	33.26	1.94	+0.57
47845.8709	0.0964	40.09	2.92	+4.81	13.83	1.88	+0.79
47896.7312	0.3151	35.14	3.12	−0.19	11.47	2.00	−1.51
47898.7686	0.4440	35.89	3.31	+4.16	16.98	2.13	+0.16
47902.6961	0.6926	19.12	3.35	−2.86	26.86	2.16	−0.38
48525.9580	0.1356	37.08	3.27	−0.22	10.20	2.10	−0.67
48670.6445	0.2920	36.83	3.41	+0.95	14.36	2.19	+1.97
48871.9572	0.0321	12.75	3.41	−7.74	23.86	2.19	−4.97
49344.8584	0.9596	−6.27	4.17	−3.48	50.94	2.69	−2.76
49373.6908	0.7842	14.82	4.56	−1.44	37.61	2.94	+4.25
49410.6311	0.1220	34.84	4.56	−2.02	7.51	2.94	−3.83
49644.8270	0.9430	3.16	3.08	+4.65	52.75	1.98	+0.43
49699.8045	0.4223	32.37	3.08	−0.02	17.20	1.98	+1.08
53693.8806	0.1867	39.09	4.82	+1.44	8.27	3.10	−2.23
56578.9704	0.7691	17.46	3.95	+0.11	34.61	3.41	+2.43
56606.9106	0.5373	22.67	3.95	−5.97	26.25	3.41	+6.12
56650.8257	0.3164	31.34	3.95	−3.96	11.49	3.41	−1.52

[a] Computed from the ephemeris given in Table 3.

Table 2
Inferred Parameters for DQ Tau

Parameter	Value	Parameter	Value
T_{10} (K)	122 ± 9	i_d (°)	162 ± 3
q	0.71 ± 0.02	φ (°)	94.0 ± 0.4
$\log M_{\rm disk}$[a] ($M_\odot$)	-3.8 ± 0.2	$v_{\rm sys}$[b] (km s^{-1})	27.15 ± 0.01
r_c (AU)	28 ± 4	δ_α ($''$)	-0.088 ± 0.003
M_* ($M_\odot$)	1.72 ± 0.40	δ_δ ($''$)	-0.246 ± 0.003
ξ (km s^{-1})	0.18 ± 0.01	d (pc)	145 ± 20

Note. — 68.3% ($\sim 1\,\sigma$) confidence intervals. We publish samples from the posterior here (Zenodo).

[a] Assuming a CO/H_2 number ratio of 7.5×10^{-5}.

[b] In the LSRK frame, for the standard radio definition. The corresponding heliocentric value is $XX \pm XX$, consistent(?) with the γ-velocity derived from the stellar orbit.

their Table 1. The 22 RV differences were combined in a weighted least-squares fit with our 33 pairs of primary/secondary velocities, yielding the elements listed in Table 3. For the individual velocities, weights were calculated from the internal errors. The LKM velocity differences were assigned reasonable nominal errors to begin with, and all uncertainties were then adjusted iteratively so as to obtain reduced χ^2 values near unity for each type of measurement (primary, secondary, RV difference). Final root-mean-square residuals, which are representative of the typical measurement errors, are given in the table. The global fit derives most of the constraint on the mass ratio from the individual primary and secondary velocities. The RV differences strongly constrain the $K_1 + K_2$ sum, but they also help to strengthen the individual K values indirectly to some extent through constraints on the remaining orbital elements. We initially allowed for a difference in the center-of-mass velocities for the primary and secondary, to account for possible biases in the RVs that may occur as a result of template mismatch, but found the difference to be insignificantly different from zero (-0.61 ± 0.71 km s^{-1}). Consequently, the final fit assumed a common value of γ.

A graphical representation of the 33 pairs of primary/secondary velocities from CfA is presented in Figure 3, along with our best-fit model from the global fit that includes the LKM velocity differences. In Figure 4 we illustrate the good agreement between the same best-fit model (solid curve) and the LKM velocity differences. The deviations between this best-fit model and a separate one that uses only the 33 individual CfA velocities (dotted line in the figure) are minimal.

Our results in Table 3 are generally consistent with those of Mathieu et al. (1997), but with uncertainties typically reduced by factors of 2–5. The minimum masses now have relative uncertainties of 5–6% instead of ∼21%.

4. DISCUSSION

We have reported our measurements using the ALMA Cycle 2 observations of DQ Tau near a wavelength of 1.3 mm, which resolve the dust and molecular line emission which orbit the binary in a circumbinary disk. We model the gas line emission in detail and measure a to-

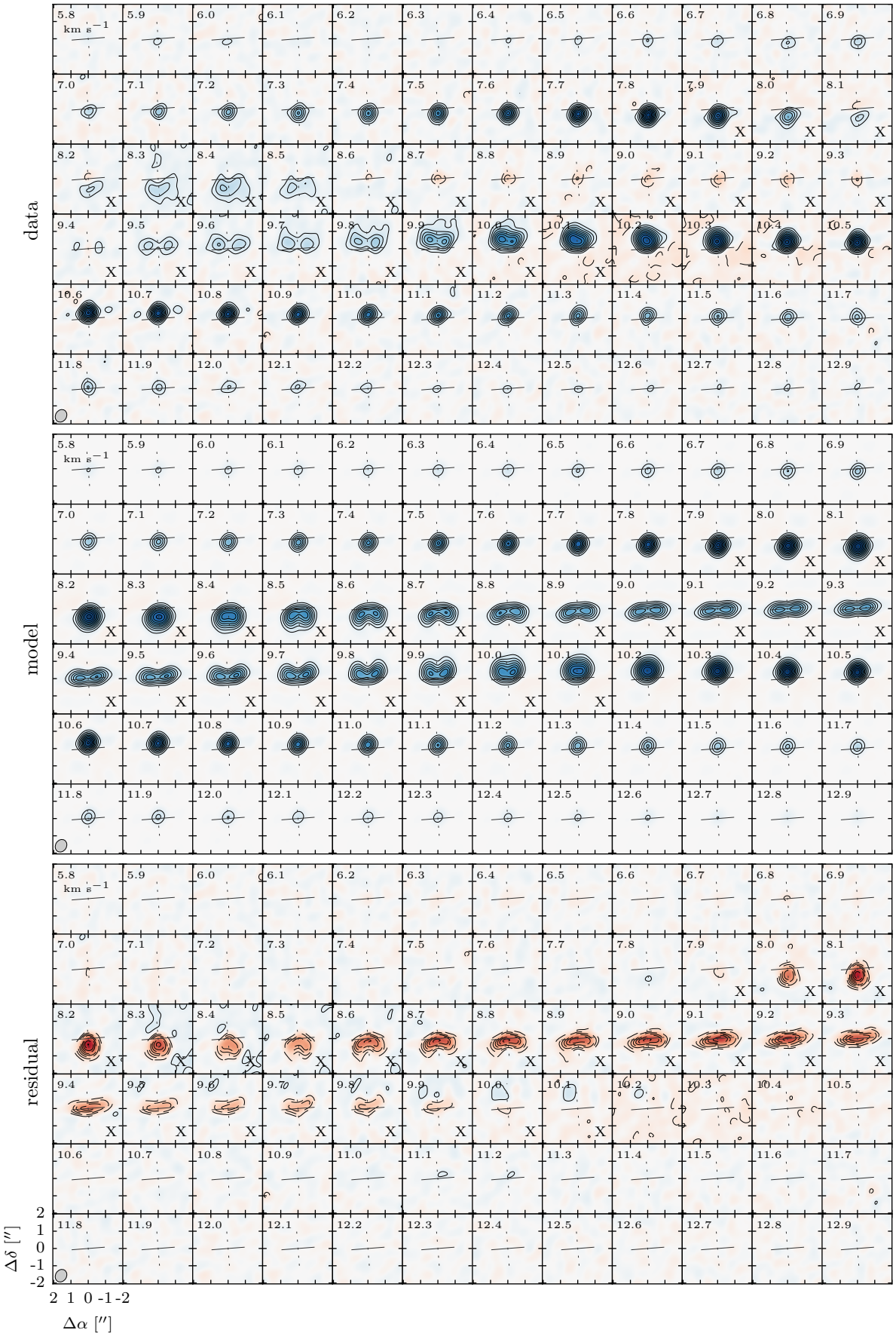

Figure 1. Continuum-subtracted CO J=2−1 channel maps for the DQ Tau data (*top*), the best-fit model (*middle*), and the imaged residuals (*bottom*) at 0.1 km s^{-1} velocity resolution. Contours are drawn at intervals of 3× the RMS noise level (9.5 mJy beam^{-1}). The synthesized beam is drawn in the lower left corner of each set of channel maps, and the LSRK velocities are labeled in each panel. We do not model the channels with $v = 7.9 - 10.1$ km s^{-1} (inclusive) because of cloud contamination; these channel maps are marked with an X.

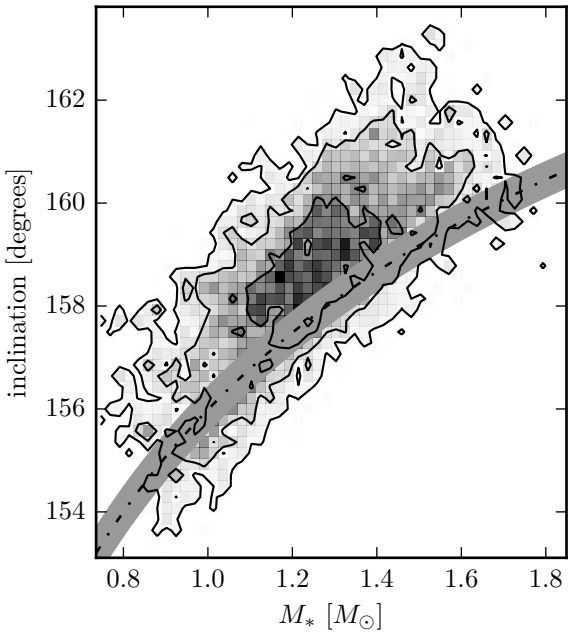

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Figure 2. *top*: the joint posterior distribution for $\{M_*,\ i_d\}$, marginalized over all other parameters. To compare with the constraints from the updated binary orbit from radial velocity monitoring measurements, we overlay ($\pm 1\ \sigma$) contours for the measurement of $M_* \sin^3 i$ (see Sect. 3.2). The disk-based mass estimate and binary radial velocity measurements would be consistent at the $1\,\sigma$ level if the disk and binary orbital planes are misaligned by by $> 2°$. *bottom*: the joint posterior distribution for $\{M_*,\ d\ \}$, which shows a linear degeneracy constrained by our distance prior.

Table 3
Updated spectroscopic orbital solution for DQ Tau.

Parameter	Value
P (days)	15.80158 ± 0.00066
γ (km s^{-1})	+24.52 ± 0.33
K_1 (km s^{-1})	20.28 ± 0.71
K_2 (km s^{-1})	21.66 ± 0.60
e	0.568 ± 0.013
ω_1 (deg)	231.9 ± 1.8
$T_{\rm peri}$ (HJD−2,400,000)	47433.507 ± 0.094
$M_1 \sin^3 i$ ($M_\odot$)	0.0348 ± 0.0017
$M_2 \sin^3 i$ ($M_\odot$)	0.0326 ± 0.0020
$(M_1 + M_2) \sin^3 i$ ($M_\odot$)	0.0674 ± 0.0033
$a_1 \sin i$ (10^6 km)	3.63 ± 0.13
$a_2 \sin i$ (10^6 km)	3.87 ± 0.11
$a \sin i$ ($R_\odot$)	10.78 ± 0.18
$q \equiv M_2/M_1$	0.936 ± 0.051
σ_1 (km s^{-1})	3.44
σ_2 (km s^{-1})	2.26
$\sigma_{\rm LKM}$ (km s^{-1})	2.50
Time span (days)	10668.8
Time span (orbital cycles)	675.2
$N_{\rm RV}$	33 × 2
$N_{\rm LKM}$	22

Note. — These results are based on a joint fit of the individual primary/secondary velocities from CfA and the LKM velocity differences. The physical constants used here are those adopted by Torres et al. (2010), consistent with the 2015 IAU Resolution B3.

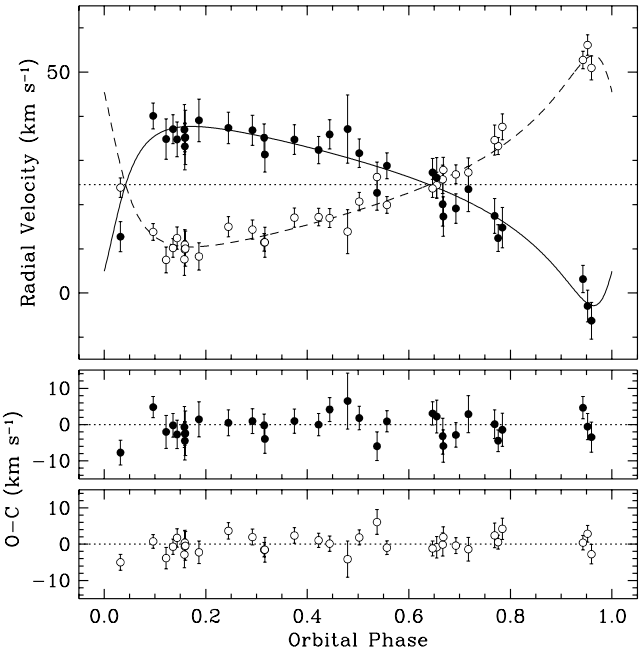

Figure 3. CfA spectroscopic observations of DQ Tau along with our best fit model that uses also the LKM velocity differences. Primary velocities are represented with filled symbols, and the dotted line marks the center-of-mass velocity. The bottom panels show the residuals.

tal mass of the central stellar binary. We first compare our measurement to previous measurements of the system using radial velocity monitoring and infrared interferometry, and then show how the mass constraint can be improved by combining these measurements into a joint posterior. Next, we follow the standard analysis of using a spectral type and SED-derived luminosity to determine a mass and age for the system via pre-main seque nce evolutionary tracks. We discuss the generally good agreement between our dynamical measurement and the predictions made by the models at these young ages and masses. Finally, we discuss pre-main sequence stellar model predictions in the context of recent dynamical mass measurements for three equal-mass bi-

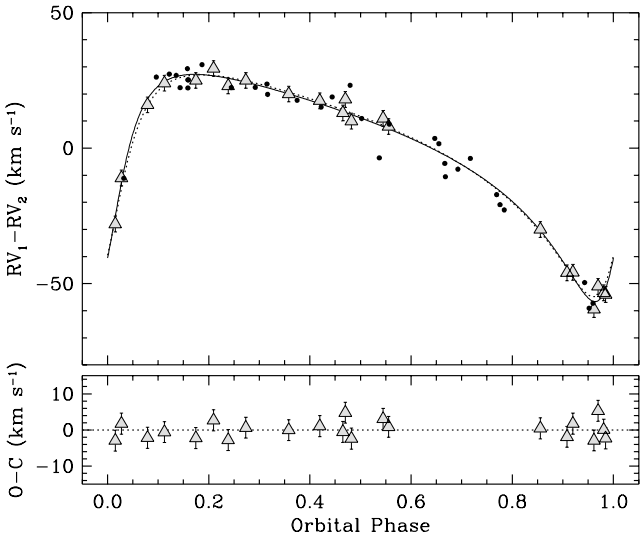

Figure 4. Predicted velocity difference according to our best-fit model (solid line), shown with the measured LKM values (triangles). Residuals are displayed at the bottom. Also shown for reference in the top panel is an orbit model that uses only the 33 individual RV measurements from CfA (dotted line), which is nearly indistinguishable from the global fit. The dots represent the velocity differences we compute from the CfA measurements, to show that both types of measurements are fully consistent with each other.

nary systems, V4046 Sgr, AK Sco, and DQ Tau.

4.1. *Comparison of Disk- and Binary-based Constraints*

Numbers in this section are generally close to final but will continue to be updated as the MCMC progresses. From our radial velocity (RV) monitoring of DQ Tau, we measure a distance-independent constraint on the total stellar mass of $M_* \sin^3 i = 0.0674 \pm 0.003\, M_\odot$, where i is the inclination of the stellar orbit, moduluo 180 degrees. By monitoring the orbit of DQ Tau with the Keck Interferometer, Boden et al. (2009) determined that the stellar orbit must have $i > 90°$, but without higher resolution observations they could not model the orbit independently of the RV measurements and so there is no precise constraint on the value of i. With high signal-to-noise, spatially resolved molecular line observations, it is possible to determine the absolute orientation of the disk inclination i_d, since there is a subtle brightness asymmetry across the plane of the disk, with the far side of the disk appearing brighter than the near side due to geometrical effects of the projected emitting area for a flared disk (Rosenfeld et al. 2012a, 2013). Unfortunately, our observations of DQ Tau do not have enough sensitivity or resolution (due to the cloud contamination of the central channels, where the emission is most extended) to determine the absolute sign of i_d. Therefore we use the Boden et al. (2009) result to constrain our disk inclination $i_d > 90°$ and use our gas measurements to conclude that $i_d = 163^{+2}_{-3}{}^\circ$.

In Figure 2, we show the joint posterior distributions for $\{M_*$, inclination$\}$ and $\{M_*$, distance$\}$. Due to the relatively face-on disk inclination, the precision of the stellar mass measurement is significantly less than an average disk-based measurement (for other near-face on examples, see CY Tau and BP Tau; Simon et al. 2000). Instead, we do precisely constrain the quantity $M_* \sin^2 i_d = 0.147 \pm 0.001\, M_\odot$, which is a consequence of modeling the projected velocity field of the disk $v_{\rm proj} = \sqrt{GM_*/r}\, \sin^2 i_d$. If we assume that the stellar orbit and the disk share the same inclination ($i = i_d$), then we can solve for the joint RV-disk constraint on the central mass and inclination, yielding $M_* = 1.2 \pm 0.05\, M_*$ and $i = 159 \pm 2°$. Finally, we can combine our distance independent measurement of M_*/d, the RV measurement of $M_* \sin^3 i_*$, and our inference on the disk inclination i_d, assuming coplanarity, to derive a dynamical distance to the system of 215 pc, which is significantly larger than our assumptions encapsulated in the distance prior.

In the context of other dynamical mass measurements such as Czekala et al. (2015) and Rosenfeld et al. (2012b), this mass measurement of DQ Tau proves that the disk-based dynamical mass technique is an effective measurement tool and validates the accuracy of our technique across a wide range of host masses. The precision of the mass measurement is strong function of the spatial and spectral resolution of the disk emission because of the degeneracy between between M_* and i_d. This means that for a near face-on disk like DQ Tau, a modest inclination uncertainty will translate into a larger uncertainty on the stellar mass, while for more edge-on disks like AK Sco, much higher precision is attainable. In the age of ALMA, the disk-based dynamical mass technique will become a workhorse technique for amassing large, statistically significant samples of young stars with accurate stellar masses, and is particularly suited to stars of earlier spectral types, given the sensitivity of ALMA. In particular, it is the *only* technique that can dynamically measure the masses of single stars.

4.2. *Comparison to Pre-MS Evolution Models*

Next, we compare our dynamical constraint on total stellar mass to the mass estimated using the traditional methods using photometry, spectroscopy, and stellar evolutionary tracks. First, we summarize the historical discussion of the spectral type of DQ Tau, which has been a source of substantial disagreement in the literature. Next, we derive a new estimate of stellar temperatures and luminosities using a two-component SED fit motivated by the non-unity mass ratio from our improved stellar orbit. We find a substantially lower reddening implied by previous fits, implying lower temperatures and luminosities than previous estimates. We use current pre-main sequence evolutionary models to map our posteriors on $T_{\rm eff}$ and L_* into posteriors on age and stellar mass and find the predicted mass is consistent with the mass estimate from our joint disk-RV analysis.

Notes about blends. All analysis has assumed equal mass ratios.

DQ Tau has a spectral type. Basri et al. (1997) determines K5, while Herbig77 determines M1. Luminosity is uncertain because of the uncertain values of extinction, ranging from $A_V = 0.5$ mag to $A_V = 2.1$ magnitudes. Luhman 03 determines M0 spectral type.

Herczeg & Hillenbrand (2014) find a spectral type of M0.6 and an $A_V = 1.40$ mag, using medium resolution optical spectra.

Bary & Petersen (2014) Based upon moderate resolution IR spectra (SPEX 0.8 - 2.4 micron and TSpec 0.9 - 2.5 micron), they find the following.

estimate spectral type using extinctions ranging from

0.97 (Kenyon + Hartmann95), 2.0 (Matheiu 97), and 2.13 (Strom 89).

Consider the shapes of molecular features like TiO, FeH, and H2O. The FeH feature strongly depends on surface gravity, with the coolest dwarf stars possessing the strongest features. The result is that many of these features were better fit a much cooler photosphere (M2.5 - M4.5). However, temperature sensitive metal lines (still in IR) such as K I and Mg I suggest better matches with M2V and M0V. Herbig 77 said DQ Tau was M1V based upon ratios of metallic lines in the optical. Basri 97 said it was M1 - 1.5V using TiO band at 7125A, but ratios of temperature sensitive Sc I lines at 6210 A suggested a K4 - 5V.

Motivated by the non-unity mass ratio from our new stellar orbit ($q = 0.93 + 0.05$), we explore a two-component fit to the spectral energy distribution (SED) assembled from the photometry from [these sources] (previously presented in Andrews et al. (2013)). We fit photometry in the B, V, R, I, and J_{2M} bands and exclude U-band and bands later than J, since these are likely contaminated by accretion activity or disk emission, respectively. We allow two effective temperatures and luminosities, but fit for the same A_V, and use the same distance prior as our disk modeling. We fix $\log g$ of both components, and run using values of 3.5, 4.0, and 4.5 dex to explore the sensitivity of this parameter. At each posterior draw, we calculate the implied mass ratio of the components and impose a Gaussian prior from the stellar orbit with $\mu = 0.93$ and $\sigma = 0.05$. Then, we use the synthetic photometry in the BT-Settl catalog of stellar models (Allard et al. 2003), interpolated to the values of $T_{\rm eff}$ and $\log g$ for solar metallicity to generate an SED and adjust for extinction using the R_V curve of XX. Similar to Czekala et al. (2015), we include a "jitter" term in each band to account for stellar variability. Because photometry alone is a poor diagnostic of effective temperature, we also adopt a prior on temperature corresponding to a range measured spectral types, $\mu = 3900$K and $\sigma = 300$K, and assert that the secondary is cooler than the primary star. The best-fit SED is shown in Figure 5, which yields convincing estimates of $T_{\rm eff}$s and L_*s, lower A_V and thus correspondingly lower $T_{\rm eff}$, lower L_*.

Next, we use our joint posterior distribution of $\{T_{\rm eff,\,0}$, L_0, $T_{\rm eff,\,0}$, $L_1\}$ and stellar evolutionary models to predict the age and masses of the components. We follow the Bayesian formalism described in Czekala et al. (2015); Rosenfeld et al. (2012b); Jørgensen & Lindegren (2005), using stellar evolutionary tracks to to jointly predict $T_{\rm eff}$ and L_* as a function of M_* and age (τ_*). For this binary fit, we assume that

Tracks produced by (Siess et al. 2000; Dotter et al. 2008; Tognelli et al. 2011; Baraffe et al. 2015). Our results are in Figure 5.

mass ratio is in agreement with RV. logg is consistent with 4.0 dex.

The weighted means of these are XX. Are the ages consistent with Taurus?

Note how binarity is important to discuss when thinking about errors. Blends in spectral types. Estimated from low resolution spectra. Show that our picture holds together from multiple angles of constraint.

Binarity and Blending are an important and confounding part of this analysis. Need to take binarity into account when doing these fits.

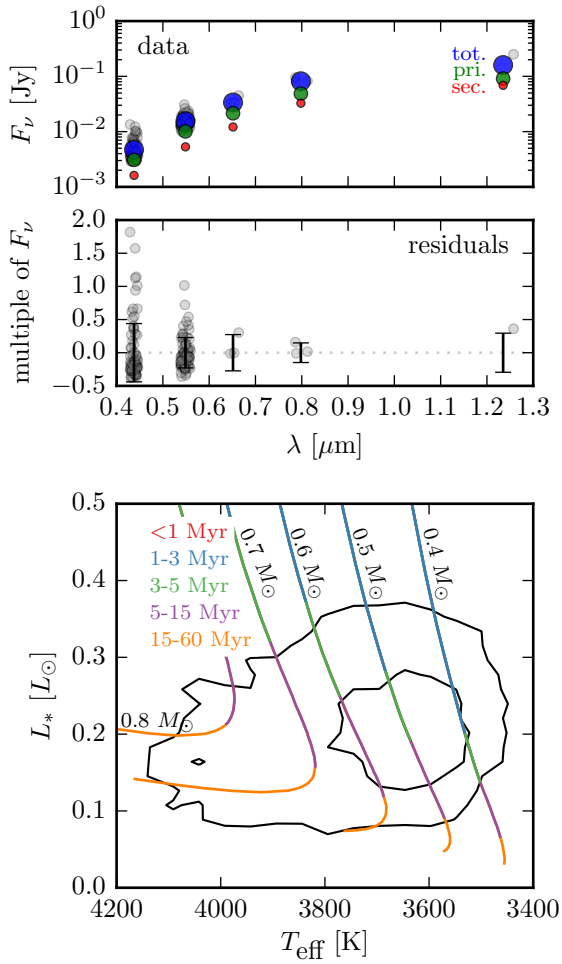

Figure 5. *top*: The H-R diagram, with the marginal posteriors inferred from the SED modeling shown as 1 and 2 σ contours. The Dotter et al. (2008) pre-MS model mass tracks are overlaid, with color scales indicating different ages (note that these values refer to each individual star in the AK Sco binary). *bottom*: The joint mass and age constraints from various pre-MS models, shown as 1 σ contours and marginalized distributions. The gray band marks the disk-based constraint on M_* (1 σ). Primary star shown.

Basri et al. (1997). Bary & Petersen (2014) Accretion activity and spots. Flares also occur near apoastron, when stars are closest to the disk. The spots have a potentially high filling factor, meaning that almost half of the photosphere would be cooler than M0. This would correspondingly reduce the luminosity of the system. This would have what sort of effect on the inferred model masses?

4.3. *Three Equal-Mass Pre-Main Sequence Binaries*

Interestingly, some of the most accessible binary systems for dynamical mass measurements have been double-lined spectroscopic binaries with near equal mass ratios.

V4046 Sgr (Rosenfeld et al. 2012b, 2013) is a close binary ($a = 0.045$ AU) with $P = 2.4$ days and a circular orbit ($e < 0.001$), a pair of K5 and K7 stars (Stempels & Gahm 2004) with mass ratio $q = 0.94$ and is

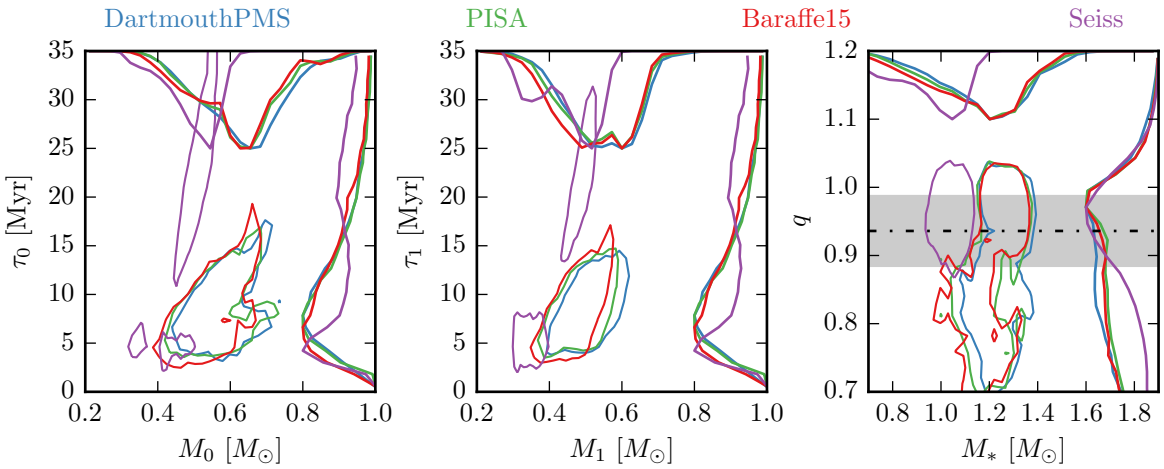

Figure 6. Using the joint posterior distribution on $\{T_{\rm eff,0}, L_0, T_{\rm eff,1}, L_1\}$ obtained from the SED fitting (Figure 5, top), we use various pre-main sequence evolutionary models to map this posterior into joint constraints on the age (τ) and mass of the two components, the stars have the same age. *left and middle panels*: joint age and mass constraints on the primary and secondary star, respectively. *right*: the joint posterior on the total system mass M_* and the mass ratio q.

Table 4
Evolutionary Model Predictions

Model Grid	M_* $[M_\odot]$	τ_* [Myr]
Siess et al. (2000)	2.70 ± 0.06	19 ± 3
Tognelli et al. (2011)	2.71 ± 0.07	18 ± 2
Dotter et al. (2008)	2.60 ± 0.07	17 ± 2
Baraffe et al. (2015)	2.68 ± 0.05	19 ± 2

Note. — The quoted uncertainties correspond to the 68.3% ($\sim 1\,\sigma$) confidence intervals.

about 10 Myr old. Coupled with the RV measurements with (Stempels & Gahm 2004), Rosenfeld et al. (2012b) determine component masses of $M_1 = 0.90 \pm 0.05 M_\odot$ and $M_2 = 0.85 \pm 0.04 M_\odot$ and an orbital inclination of $i = 33.42 \pm 0.01°$.

AK Sco is a pair of near-identical F5 stars ($q = 0.99$) (Alencar et al. 2003) with $P = 13.6$ days on eccentric orbits $e = 0.47$. Czekala et al. (2015) determine a total mass for the system, which coupled with the near-equal mass ratio, translates to component masses $M_1 \approx M_2 = 1.25 \pm 0.05 M_\odot$.

As we presented in Sec 4.1, DQ Tau is a pair of M0-1 stars with mass ratio $q = 0.97$. Our measurement translates to $M_1 = XX \pm XX M_\odot$ and $M_2 = XX \pm XX M_\odot$.

Can we say anything about the theories of Tokovinin (2000); Bate (2000) for accretion equalization from the disk given that we now have imaged the disk, know the disk mass and size, and the age of the system?

Discussion of the behaviour, is this higher or lower. Older models with higher masses tend to do quite well, but the younger, cooler stars suffer. While the older models tend to do well, there is modest tension for AK Sco.

5. SUMMARY AND CONCLUSIONS

compact disk with mass XX detected in CO and ^{13}CO J=2−1 transitions.
use parametric disk structure to forward model ALMA visibilities to measure mass
updated orbital elements, including significant improvement in uncertainties

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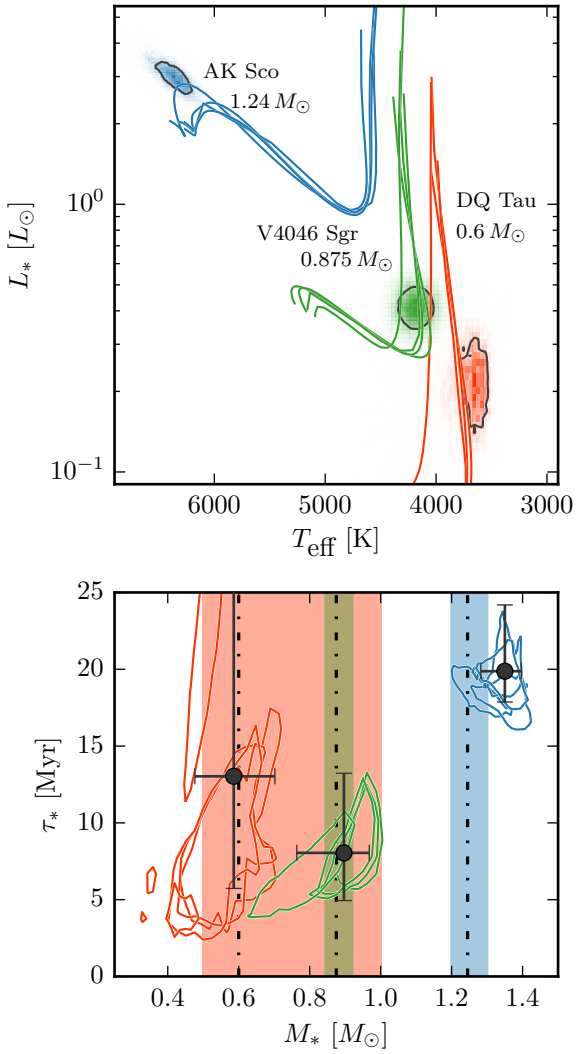

Figure 7. *top*: HR diagram of three equal mass, double-lined spectroscopic binaries. *bottom*: translated into mass and age constraints (error bars show 16 to 84% confidence levels), overlaid with our dynamically measured masses.

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